

BUSINESS SALES PERFORMANCE ANALYTICS

A Power BI Business Analytics & Data Visualization Project

Future Interns – Data Science & Analytics Internship

Task 1

Prepared by: Saniya Kalawant

Project Overview

The objective of this project is to analyse business sales data and build a professional, client-ready Power BI dashboard to understand sales performance, profitability, product performance, and regional performance. The dashboard consolidates transactional sales data into an interactive analytical tool that supports data-driven business decisions.

Objectives

- Clean and organize business sales data
- Analyze sales and revenue trends
- Identify top-performing products and sub-categories
- Compare category and regional performance
- Analyze profitability
- Identify loss-making areas
- Build an interactive Power BI dashboard
- Generate actionable business recommendations

Tools & Technologies

- Power BI
- Power Query
- DAX
- Data Visualization
- Business Analytics

Dataset, Data Preparation & KPIs

Dataset

The project uses a Superstore sales dataset containing transactional information such as Order Date, Sales, Profit, Category, Sub-Category, Product, Customer and Region.

Data Preparation

- Checked duplicate records
- Reviewed missing values
- Verified data types
- Organized date fields
- Created a Date Table for time-based analysis
- Used Power Query for data transformation
- Created DAX measures for KPIs and analysis

Key Performance Indicators (KPIs)

KPI	Value
Total Sales	\$2.30M
Total Profit	\$286.40K
Profit Margin	12.47%
YoY Sales Growth	46.88%

Additional KPIs tracked on the dashboard include:

- Latest Year Sales
- Total Orders
- Total Quantity
- Average Order Value

Dashboard Overview

Dashboard Home

The Dashboard Home page is a navigation-focused landing page providing access to the major analytical sections of the report: Executive Overview, Product Analysis, and Regional & Customer Analysis.



Figure 1: Dashboard Home

Executive Overview

The Executive Overview page provides a high-level view of Sales, Total Profit, Profit Margin, Total Orders, Total Quantity, and Average Order Value, along with sales trends, category performance, regional performance, and top-performing products.



Figure 2: Executive Overview

Product & Regional Analysis

Product Analysis

This section evaluates sales and profitability across products, sub-categories and categories, and identifies high- and low-performing areas.



Figure 3: Product Analysis

Regional & Customer Analysis

This section provides regional and customer-focused analysis to identify performance patterns and business opportunities.



Figure 4: Regional & Customer Analysis

Key Insights, Risks & Recommendations

Key Insights

■ Technology – Top Performing Category

Technology generated approximately \$836.15K in sales and achieved the highest profit margin of approximately 17.40%.

■ Furniture – Lower Profitability

Furniture generated approximately \$742.00K in sales but showed comparatively lower profitability.

■ Phones – Highest-Selling Sub-Category

Phones generated approximately \$330K in sales.

■ Copiers – Highest Profit

Copiers generated approximately \$56K in profit.

■ Tables – Major Profitability Concern

Tables generated approximately \$207K in sales but resulted in approximately \$18K loss.

Business Risks

- Tables are loss-making despite high sales.
- Bookcases and Supplies also generate negative profits.
- Furniture has comparatively lower profitability.
- High sales volume does not always translate into high profitability.

Growth Opportunities

- Focus growth efforts on Technology.
- Promote high-profit sub-categories such as Copiers, Phones and Accessories.
- Cross-sell profitable products with high-selling products.
- Improve Furniture profitability through pricing and product-mix optimization.
- Investigate loss-making sub-categories.

Recommended Actions

1. Review pricing, discounts and costs for Tables.
2. Increase focus on high-margin Technology products.
3. Optimize Furniture's product mix and pricing.
4. Promote profitable products such as Copiers and Accessories.
5. Monitor profit margin alongside sales.

Conclusion

This analysis demonstrates that overall sales performance is strong, with steady year-over-year growth and a healthy profit margin at the business level. However, profitability varies significantly across categories and sub-categories: Technology stands out as the strongest performer, while certain sub-categories such as Tables, Bookcases and Supplies erode profitability despite contributing to sales volume. The interactive Power BI dashboard consolidates these insights into a single tool, enabling stakeholders to make informed, data-driven decisions around pricing, product mix and regional strategy.

Note: The Power BI .pbix file could not be included due to a technical file corruption issue. Dashboard screenshots and project documentation are provided as evidence of the completed analysis.